

Optimizing Predictive Models for Drug-Induced Suicidal Risk: A Data-Balanced Framework with Explainable AI

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Abstract—A thorough investigation is needed to understand the rising global patterns of drug addiction and suicide rates. This study explores the link between drug addiction and suicidal ideation to develop a prediction model for early detection and support of at-risk individuals. An exhaustive analysis employs computational complexity, performance metrics, AUC-ROC curves, and cross-entropy loss to meticulously assess and compare the efficacy by using a suite of machine learning algorithms, such as Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN), and Extra Trees Classifiers on imbalanced and balanced datasets. By implementing advanced techniques like SMOTE for data balancing, the reliability and precision of predictions are ensured. The Extra Trees Classifier, with hyper-parameter tuning, is the most effective model, achieving a 94% accuracy rate, 0.95 AUC score, and 0.2589 cross-entropy loss. The study emphasizes explainable AI, providing logical insights and detailed explanations for model selection across different parameters. This thorough analysis ensures transparency and fosters trust in the predictive capabilities of models. The findings set new benchmarks in predictive modeling and intervention strategies, with implications for clinical practices and public health strategies. The potential to transform intervention methods and save lives underscores the critical importance of this study.

Index Terms—Explainable AI (XAI), Drug Addiction, Suicidal Tendencies, Data Imbalance, SMOTE, AUC-ROC, Cross-Entropy Loss, Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors, Extra Trees Classifier, Hyperparameter Tuning, Accuracy, Computational Complexity,

beneficial model. The research endeavors to provide insight into the reasons behind the different behaviors exhibited by these models, ultimately aiming to improve the prediction of suicide risks in drug-addicted individuals [1].

From our study, we are poised to contribute three state-of-the-art advancements to the field:

- 1) **Enhanced Understanding of Drug-Induced Suicidal Risks:** The research creates a model linking drug addiction with suicide risk, allowing for early interventions and potentially saving lives.
- 2) **Addressing Data Imbalance:** The study uses advanced data balancing techniques like SMOTE to address imbalanced healthcare datasets, ensuring accurate model predictions in real-world scenarios.
- 3) **Explainable Algorithm Performance:** Use of Explainable AI to analyze the results and select the best model for real-world healthcare scenarios, providing actionable insights and reasons for excluding other algorithms.

These contributions advance predictive modeling and suicide prevention in both clinical and public health domains.

The rest of this paper is arranged as follows: section II gives a detailed summary of previous research, section III examines the methodology and section IV focuses on the findings with appropriate elaboration. Finally, in section V, the conclusion was drawn.

I. INTRODUCTION

Suicide and drug addiction are closely linked. This research employs machine learning to predict suicidal tendencies associated with drug addiction, enabling timely interventions [2]. Accurate predictions are vital [3], and recent advancements emphasize the need for explainable machine learning models in mental health, addressing imbalanced datasets and ensuring ethical practices in healthcare. This study delves into the application of various machine learning models, considering factors such as time complexity, performance metrics, AUC-ROC curve, and cross-entropy loss [8]. Additionally, hyperparameter tuning is conducted using preprocessed datasets for both imbalanced and balanced datasets [15] to select the most

II. RELATED WORKS

Chhetri et al. [1] examine ML models for understanding substance use disorders through brain imaging and behavior studies. Ehtemam et al. [2] highlight key risk factors like age and mental health, urging further research on ethical concerns in ML integration. Ribana Roscher et al. [3] stress the importance of transparent predictive modeling in addressing drug addiction and suicide. Maonan Wang et al. [4] introduce a machine learning framework for cybersecurity, while Yi-Wei Ma et al. [5] focus on explainable learning in early Parkinson's diagnosis. In education, Khosravi et al. [6] propose tools using explainable ML, and Kani et al. [7] emphasize its healthcare applications for precision and efficiency.

In drug relapse prediction, Xiong et al. [8] demonstrated that SMOTE and ROSE techniques improve model performance with imbalanced datasets. Ramesh et al. [9] and Nguyen et al. [10] also showed the effectiveness of machine learning in detecting heart disease and diabetes. Tasnim et al. [11] compared Decision Tree, Support Vector Machine, and Naive Bayes across seven datasets, emphasizing model explainability and performance metrics for algorithm selection.

In addition research, Liu et al. [12] and Almasi et al. [13] created predictive models for drug abuse, while Rahman et al. [14] and Li et al. [15] examined mental health aspects, emphasizing machine learning's role in understanding addiction behaviors. The link between drug addiction and suicidal behavior is increasingly studied using machine learning for better predictions. Maher Maalouf [16] used logistic regression for drug addiction predictions, and Dragisic et al. [17] highlighted the elevated suicide risk in drug addicts. Additionally, Islamova [18] applied logistic regression to identify drug use risk among boys in Russia to facilitate early detection. In summary, these studies emphasize the progress in machine learning for predicting drug addiction and suicidal behavior, stressing the importance of accurate, interpretable, and adaptable models across various datasets and uses.

III. METHODOLOGY

This section outlines the methods used to predict suicidal tendencies influenced by drug addiction using machine learning algorithms.

A. Algorithms Evaluated

We evaluated the following algorithms:

- Logistic Regression (LR)
- Random Forest (RF)
- Gradient Boosting (GB)
- K-Nearest Neighbors (KNN)
- Extra Trees Classifier (ETC)

1) *Logistic Regression*: A linear classifier that models the probability of a binary outcome using a logistic function.

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}}$$

Where $P(Y = 1|X)$ represents the probability, and $\beta_0, \beta_1, \dots, \beta_n$ are coefficients for input features.

2) *Random Forest*: An ensemble method that aggregates multiple decision trees to reduce overfitting and improve accuracy.

$$f(x) = \frac{1}{T} \sum_{t=1}^T f_t(x)$$

Where $f_t(x)$ is the prediction from the t -th tree.

3) *Gradient Boosting*: Builds models sequentially, with each model correcting errors from the previous one using gradient descent.

$$F_m(x) = F_{m-1}(x) + h_m(x)$$

Where $F_m(x)$ is the model at step m , and $h_m(x)$ corrects errors.

4) *K-Nearest Neighbors (KNN)*: Classifies an instance based on the majority class of its k -nearest neighbors in the feature space.

$$\hat{y} = \frac{1}{k} \sum_{i \in N_k(x)} y_i$$

Where $N_k(x)$ are the k closest neighbors.

5) *Extra Trees Classifier*: Similar to RF, but introduces more randomness when splitting nodes, leading to greater diversity among trees.

$$f(x) = \frac{1}{T} \sum_{t=1}^T f_t(x)$$

B. Hyperparameter Tuning

Purpose: Optimize algorithm performance by adjusting key parameters like a number of trees, maximum depth, and minimum samples per split.

Process:

- **Parameter Grid:** Define a range of hyperparameters.
- **Cross-validation:** Use multiple data folds to assess each combination of hyperparameters.
- **Best Parameters:** Select the parameters that yield the best cross-validation performance.

C. Parameters for Result Analysis

We focus on several key metrics for evaluating model performance:

1) *Computational Complexity*: Understanding computational complexity is crucial for assessing the feasibility of algorithms, particularly with different datasets. Here are the complexities, in Table I.

- n = number of samples
- d = number of features
- m = number of trees
- k = number of boosting stages

2) *AUC-ROC Curve*: The AUC-ROC curve measures classification performance. A higher AUC indicates better model ability to distinguish between classes:

$$\text{AUC} = \int_0^1 \text{TPR}(t) d\text{FPR}(t)$$

3) *Stratified K-Fold Cross-validation*: This method ensures robust model evaluation by partitioning the data into K folds, preserving class proportions:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

TABLE I: Computational Complexity of Machine Learning Algorithms.

Algorithm	Training Time	Prediction Time	Training Space	Prediction Space
Logistic Regression	$O(n \cdot d^2)$	$O(d)$	$O(d^2)$	$O(d)$
Random Forest	$O(m \cdot n \cdot d \cdot \log n)$	$O(m \cdot d \cdot \log n)$	$O(m \cdot d \cdot \log n)$	$O(m \cdot d)$
Gradient Boosting	$O(k \cdot m \cdot n \cdot d)$	$O(m \cdot d \cdot \log n)$	$O(m \cdot d \cdot k)$	$O(m \cdot d)$
K-Nearest Neighbors (KNN)	$O(1)$	$O(n \cdot d)$	$O(n \cdot d)$	$O(n \cdot d)$
Extra Trees Classifier	$O(m \cdot n \cdot d \cdot \log n)$	$O(m \cdot d \cdot \log n)$	$O(m \cdot d \cdot \log n)$	$O(m \cdot d)$

4) *Cross Entropy Loss*: This loss function measures the difference between predicted probabilities and true labels:

$$\text{Cross Entropy Loss} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)]$$

For multi-class classification:

$$\text{Cross Entropy Loss} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c})$$

D. Workflow Diagram

Workflow for Research Project [Figure 1](#) on Predicting Suicidal Tendencies Influenced by Drug Addiction Using Machine Learning Algorithms through Different Parameters.

IV. RESULTS AND DISCUSSIONS

This section provides a synopsis of the datasets, pre-processing, balancing method, experimental results overview, and conclusions drawn from the data.

1) **Dataset Overview**: The dataset from Kaggle examines the link between drug addiction and suicidal thoughts in Bangladesh. It includes 437 data points and 23 features. The main goal is to identify predictors of suicidal thoughts, with the target variable being a binary indicator of the presence or absence of suicidal thoughts among respondents.

2) Dataset Pre-processing:

- **Exploratory Data Analysis (EDA)**: This analysis is essential for deeply understanding the dataset's structure, distributions, and feature relationships. The Correlation Matrix in [Figure 2](#) shows relationships between various variables, highlighting both positive and negative correlations. [Figure 3](#) presents the distribution of suicidal thoughts across categories of ease of drug control and frequency of drug usage.
- **Data Cleaning**: Involved in handling missing values through imputation techniques and identifying/treating outliers using Z-scores.
- **Data Transformation**: Categorical variables were transformed into numerical formats using label encoding.

3) **Dataset Balancing**: The techniques were evaluated to address class imbalance and improve model performance, among the methods tested—under-sampling, over-sampling, and SMOTE (Synthetic Minority Oversampling Technique) emerged as the most effective. It significantly improved the model's accuracy, precision, recall, and F1-Score. The following [Table II](#) summarizes the performance metrics for each technique:

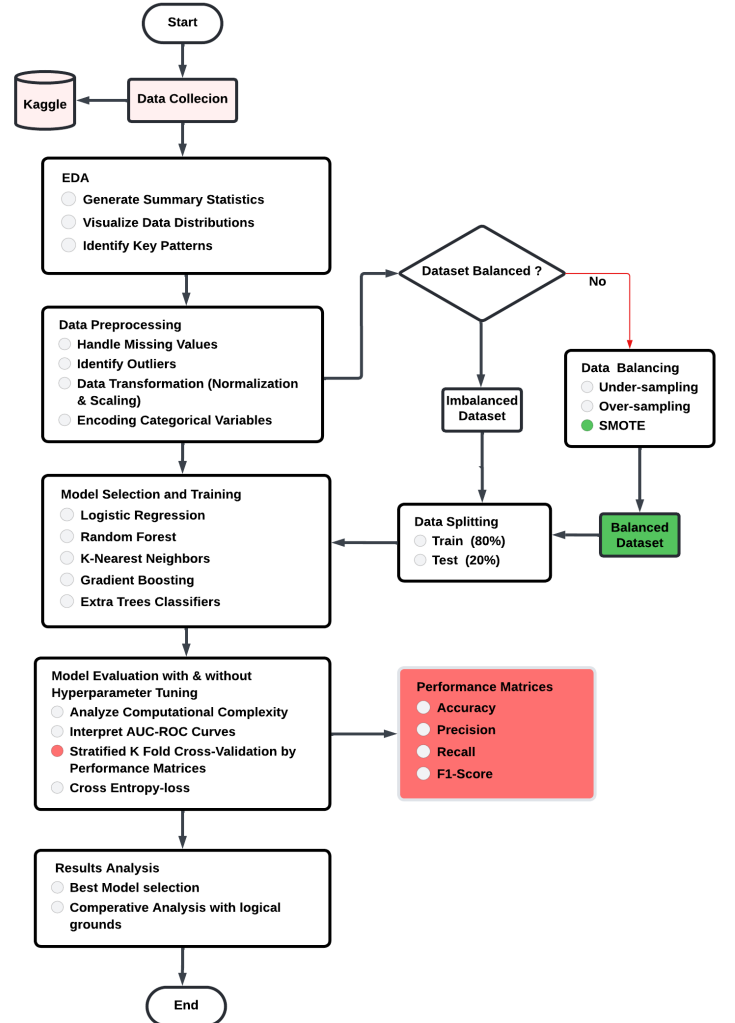


Fig. 1: Workflow for Research Project

TABLE II: Performance Metrics for Balancing Techniques

Technique	Accuracy	Precision	Recall	F1-Score
Under-sampling	0.82	0.78	0.80	0.79
Over-sampling	0.86	0.81	0.78	0.79
SMOTE	0.88	0.83	0.82	0.82

A. Experimental Results

The experiment has been conducted based on Comparative analysis with four contexts as illustrated below:

- 1) Exp. A: Computational Complexity.

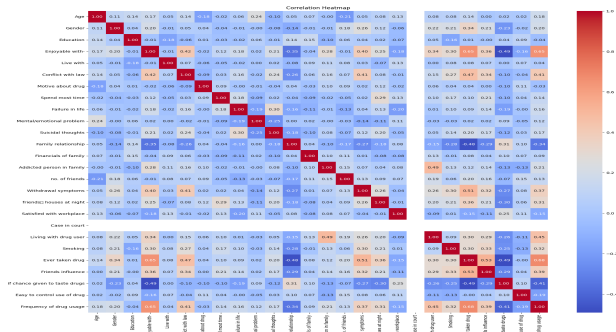


Fig. 2: Correlation Matrix

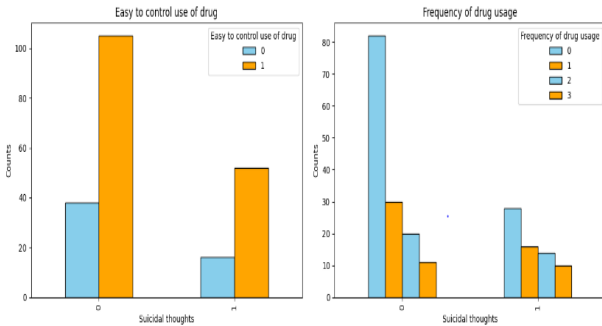


Fig. 3: Bar Plot

- 2) Exp. B: ROC-AUC Analysis.
- 3) Exp. C: Stratified K-Fold Cross-Validation by Performance Matrices.
- 4) Exp. D: Cross-Entropy Loss.

B. Result of Exp. A

Overall Calculation of Complexity: Dataset Parameters:

- Number of Training Examples (n): 437
- Number of Features (d): 23

The analysis Table III compares ML algorithms for suicide prediction. LR is efficient for real-time predictions. GB has high training complexity but offers robust performance. RF and ETC are effective but may not be ideal for limited computational power.

C. Result of Exp. B

AUC-ROC Curve for both Imbalanced and Balanced Dataset: Figures 4 and 5, illustrate how effectively each model distinguishes between classes. Table IV quantifies this performance by presenting the AUC scores, confirming that models generally achieve higher performance on the balanced dataset than the imbalanced one. This suggests that balancing the dataset improves the models' ability to correctly identify positive and negative cases.

D. Result of Exp. C

Stratified K-Fold Cross-Validation by Performance Metrics:

- **IDMP**: Imbalanced Dataset Model Prediction
- **HTID**: Hyperparameter Tuning on Imbalanced Dataset
- **BDMP**: Balanced Dataset Model Prediction
- **HTBD**: Hyperparameter Tuning on Balanced Dataset

The performance metrics in Table V In IDMP, ensemble methods like RF and GB performed well, handling the class imbalance effectively and achieving higher precision and recall. Simpler models like LR and KNN struggled with performance due to imbalance. In HTID, hyperparameter tuning improved accuracy for all algorithms, but ensemble methods like RF and ETC outperformed others. In BDMP, balancing the dataset enhanced model performance, particularly for RF and ETC, while LR and GB also improved. In HTBD, further tuning highlighted ETC as the top performer, underscoring the significance of balancing and tuning in predictive modeling. Performance Metrics Mean Table:

Explanation of Shift in Performance:

- **Random Forest**: Outperformed on imbalanced data due to robust ensemble methods, but exhibited a slight bias towards the majority class.
- **Extra Trees Classifier**: Performed better on balanced data as the increased randomness in splits reduced overfitting and improved generalization.

1) **Result of Exp. D**: The cross-entropy analysis Table VI shows that balancing the dataset improves most models, with RF, ETC benefiting the most. LR remains consistent, while KNN shows an increased loss on the balanced dataset, indicating its inadequacy for this task. Overall, balancing the dataset enhances model performance, with the ETC being the most robust in reducing cross-entropy loss

E. Discussion

This Table VII illustrates the significance of Explainable AI by highlighting how diverse parameters across various algorithms reveal performance disparities and guide informed model selection.

1. Impact of Algorithm Selection on Model Performance

- **Computational Complexity**: LR was the most computationally efficient, making it suitable for real-time applications. However, more complex models like RF and ETC offered better predictive accuracy, though at the cost of increased computational demand.
- **AUC-ROC & Cross-Entropy Loss**: ETC and RF achieved an AUC-ROC of 0.95 on the balanced dataset, indicating a strong ability to distinguish between classes. Additionally, ETC demonstrated substantial improvements in cross-entropy loss, dropping from 0.5213 (on the imbalanced dataset) to 0.2589 (on the balanced dataset).
- **Performance Metrics**: The ETC algorithm achieved the highest performance metrics in terms of accuracy (94%) and F1-score on the balanced dataset, showing its effectiveness in handling both class imbalance and generalizing to unseen data.

2. Handling Class Imbalance with SMOTE

TABLE III: Complexity Analysis of Various Algorithms

Algorithm	Training Time Complexity	Prediction Time Complexity	Training Space Complexity	Prediction Space Complexity
Logistic Regression	$O(231173)$	$O(23)$	$O(529)$	$O(23)$
Random Forest	$O(21011158)$	$O(48080)$	$O(48080)$	$O(2300)$
Gradient Boosting	$O(10051000)$	$O(48080)$	$O(23000)$	$O(2300)$
K-Nearest Neighbors (KNN)	$O(1)$	$O(10051)$	$O(10051)$	$O(10051)$
Extra Trees Classifier	$O(21011158)$	$O(48080)$	$O(48080)$	$O(2300)$

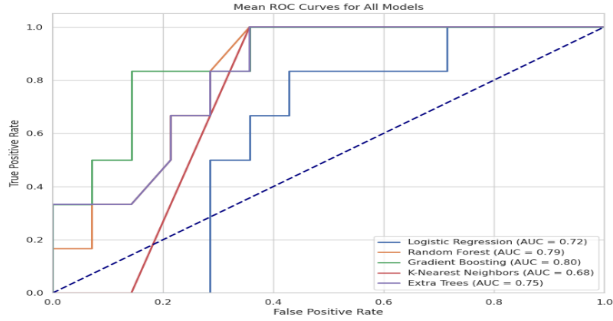


Fig. 4: Mean ROC Curves for All Models (Imbalanced Dataset)

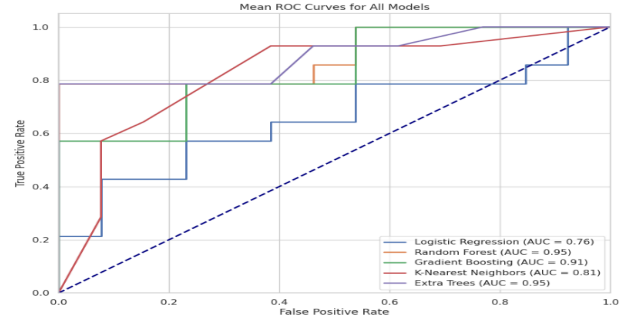


Fig. 5: Mean ROC Curves for All Models (Balanced Dataset)

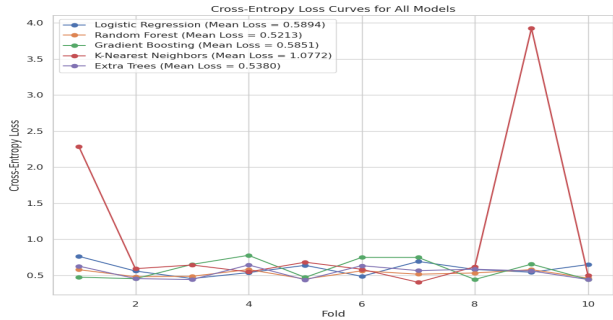


Fig. 6: Cross-Entropy Loss Curves for All Models on Imbalanced Dataset



Fig. 7: Cross-Entropy Loss Curves for All Models on Balanced Dataset

TABLE IV: AUC-ROC Scores for Different Models

Model	Imbalanced Dataset	Balanced Dataset
Logistic Regression	0.72	0.76
Random Forest	0.79	0.95
Gradient Boosting	0.80	0.91
K-Nearest Neighbors	0.68	0.81
Extra Trees	0.75	0.95

- The improvements in precision and recall metrics after balancing demonstrated the positive impact of SMOTE in handling imbalanced classes, ensuring that the model captures both the minority (suicide risk) and majority (non-suicide risk) classes more effectively.

3. Hyperparameter Tuning and Performance Optimization

- Extra Trees, after tuning, provided the best overall performance with 94% accuracy, outperforming other models.
- RF and GB, though requiring higher computational costs, also benefitted from hyperparameter tuning, which improved their accuracy and reduced overfitting.

TABLE V: Performance Metrics Mean Table

Algorithm	Section	Accuracy	Precision	Recall	F1-Score
Logistic Regression	IDMP	0.68	0.70	0.70	0.69
	HTID	0.69	0.67	0.69	0.67
	BDMP	0.73	0.75	0.73	0.72
	HTBD	0.85	0.74	0.73	0.73
Random Forest	IDMP	0.73	0.74	0.73	0.70
	HTID	0.75	0.76	0.75	0.72
	BDMP	0.86	0.87	0.85	0.85
	HTBD	0.87	0.88	0.87	0.87
Gradient Boosting	IDMP	0.72	0.73	0.72	0.71
	HTID	0.73	0.75	0.73	0.72
	BDMP	0.84	0.85	0.84	0.84
	HTBD	0.86	0.87	0.86	0.86
K-Nearest Neighbors	IDMP	0.68	0.65	0.68	0.62
	HTID	0.69	0.65	0.69	0.62
	BDMP	0.64	0.65	0.64	0.63
	HTBD	0.80	0.82	0.80	0.79
Extra Trees Classifier	IDMP	0.71	0.71	0.70	0.71
	HTID	0.74	0.75	0.74	0.71
	BDMP	0.91	0.90	0.88	0.88
	HTBD	0.94	0.91	0.90	0.91

- Hyperparameter tuning is crucial for enhancing model performance and ensuring the model fits the dataset's characteristics.

TABLE VI: Cross-Entropy Loss Scores

Model	Imbalanced Dataset	Balanced Dataset
Logistic Regression	0.5894	0.6227
Random Forest	0.5213	0.3155
Gradient Boosting	0.5851	0.4034
K-Nearest Neighbors	1.0772	1.4850
Extra Trees Classifier	0.5380	0.2589

TABLE VII: Comparison of Algorithms Across Parameters

Parameter	Imbalanced Dataset	Balanced Dataset
Computational Complexity	LR (Low)	LR (Low)
AUC-ROC	GB (0.80)	RF (0.95) Extra Trees (0.95)
Performance Metrics	RF (0.75)	Extra Trees (0.94)
Cross Entropy Loss	Extra Trees (0.75)	Extra Trees (0.2589)

4. Generalizability Across Data Splits

- Reliable generalization across diverse subsets of the data. The Extra Trees model, after being balanced with SMOTE and fine-tuned, achieved 94% accuracy, demonstrating its ability to generalize well to different population subsets.
- This generalizability prevents overfitting, allowing the model to predict suicide risk accurately across diverse demographic groups in real-world applications.

5. Explainable AI (XAI) for Transparency and Trust

- XAI clarified why Extra Trees outperformed other models in accuracy and reduced cross-entropy loss, emphasizing the role of SMOTE in improving predictions for imbalanced data.
- The transparency offered by XAI ensures that clinical professionals can trust the model's predictions, as they can clearly see how each model and its parameters impact the outcomes. This is critical in healthcare settings, where understanding model behavior is essential before applying predictions for interventions.

V. CONCLUSION

The study highlights the crucial connection between drug addiction and suicidal behavior, underscoring the necessity for precise predictive models in mental health. Using advanced machine learning techniques and data balancing methods like SMOTE, we've created a framework that improves prediction accuracy and ensures fair representation of at-risk populations. Our findings show that the fine-tuned Extra Trees Classifier achieved remarkable performance, with a 94% accuracy rate and a 0.95 AUC, along with a cross-entropy loss score of 0.2589, also considering computational complexity. Emphasizing XAI enhances transparency and trust in predictive models, allowing for practical clinical applications and better mental health outcomes. This research establishes new standards in predictive modeling for mental health and opens pathways for future studies on algorithmic transparency and ethical aspects of machine learning. Future work will expand the dataset, incorporate additional features, and explore their impact on model performance and predictive capacity. This will enhance

the robustness of the model and enable the development of real-time healthcare interventions. Additionally, efforts will be made to ensure ethical AI use, collaborate with mental health professionals, and refine the model to prevent suicidal behavior related to drug addiction. These advancements will help improve predictive capabilities and provide more accurate insights for timely interventions.

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